Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Resource-efficient Real-time Quick Anomaly Detection Technology			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity (√), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Knowledge Services		Knowledge Services	
	Medium Category	Small Category	Medium Category	Small Category
	R&D / Engineering Services	Product Quality Management Technologies	R&D / Engineering Services	Testing / Inspection / Analysis Techniques
Concept of Proposed Technology	The proposed technology aims to develop a real-time quick anomaly detection technology that quickly identify, localize, and react to changes in a complex system with optimal balance between detection delay and resource utilization. The system leverages advanced artificial intelligence and systems informatics to enable automated detection infrastructure modeling and design, as well as real-time operations software. Key features: • Automated modeling of complex systems (e.g., production system, IoT infrastructure)			

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	• Machine learning based analysis for the detection performance (e.g., detection delay) of the system given resource constraints (e.g., inspection cost, computational power) • Using advanced adaptive sampling techniques, we expect a reduction of ~93% detection cost while keeping detection performance within 10% variations • Optimal design of anomaly detection framework based on cost analysis • Software to display, analyze, and monitor real-time data • Quick detection of anomalies and large language model-based recommendations to react to the anomalies
Rationale for Joint R&D Project	The proposed project requires close collaboration between a research institute and an industrial partner to successfully translate modeling, monitoring, and AI technologies into a practical anomaly detection system. A joint R&D framework is essential to integrate advanced analytical algorithms with real industrial environment and requirement and to validate the technology under realistic conditions. Rationale for collaboration: • The research institute provides expertise in advanced analytics techniques • The Korean SME partner provides capabilities in sensing framework, system integration, and real-world experimentation • Effective co-design of detection software and hardware implementation is necessary to validate the efficiency of anomaly detection Expected benefits of joint R&D: • Accelerated transition from data analytics research to deployable software prototypes • Reduced technical risk through iterative hardware–software co-development and validation • Creation of a cost-effective foundation for commercialization of anomaly detection technology
Key Words	1. Anomaly detection 2. Adaptive sampling 3. Artificial intelligence 4. Risk identification and mitigation 5. Large language models

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) [2]	Start of the Project	Level (3)	End of the Project	Level (6)
R&D Objectives	The objective of this R&D project is to develop a real-time quick anomaly detection technology that quickly identify, localize, and react to changes in a complex system with optimal balance between detection delay and resource utilization. The specific R&D objectives are: • To develop AI-based detection algorithms that enable quick and highly reliable anomaly detection in engineering settings • To develop adaptive sampling-based monitoring resource allocation algorithms to optimize monitoring resources for the detection tasks given detection requirements • To develop large language model-based decision support systems that generate effective recommendations to address anomalies detected • To develop an end-to-end software prototype for the abovementioned functions			
R&D Target Specifications	The target performance specifications of this R&D project are as follows: **Detection Performance (>90%):** Achieve an anomaly detection success rate greater than 0.99 in industrial environments, and an average detection speed of more than 90% of the theoretically optimal speed. **Reduction in monitoring resource costs (90%):** Achieve a 90% reduction of monitoring costs (e.g., data acquisition cost, inspection cost, computation cost, data storage cost) while achieving the detection performance goals. **Anomaly identification performance (80%):** Achieve an anomaly identification success rate of at least 80% for common anomalies, and generate mitigation recommendations with a correct rate of at least 80%.			

R&D Plan	This R&D project will be conducted over three years, following a progressive development and validation strategy from simulation to laboratory environments and finally to real-world industrial settings.
	
	• Year 1: Simulation-Based Development and Core AI Technology Implementation
	  ○ Establish the modeling of the real use cases
	  ○ Develop an integrated simulation environment for the detection tasks
	  ○ Implement AI-based anomaly detection methods
	  ○ Implement Adaptive sampling-based resource allocation methods
	  ○ Validate core algorithms through simulation-based experiments and performance benchmarking
	• Year 2: Laboratory-Scale Integration and real case studies
	  ○ Integrate anomaly identification in a laboratory environment
	  ○ Develop large language model-based user interfaces for decision support
	  ○ Customize and optimize AI algorithms for real hardware constraints and sensor noise
	  ○ Conduct repeated laboratory experiments to evaluate robustness, safety, and task success rates			

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	○ Test on real industrial data ● Year 3: Real-World Deployment and Technology Advancement ○ Deploy the integrated system in realistic industrial environments ○ Perform large-scale experiments to verify detection, resource optimization, and anomaly mitigation performances ○ Refine the systems based on experimental results Assess commercialization potential and prepare the technology for industrial transfer

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The research institute will be responsible for the development of core AI and adaptive sampling algorithms for the proposed anomaly detection technology. The institute will lead the research on perception, decision-making, and control methodologies required to achieve the performance target in industrial environments. The specific roles and responsibilities of the research institute include: ● Development of AI-based anomaly detection algorithms, adaptive sampling algorithms, and large language models in the project ● Design of the overall system architecture and integration framework ● Simulation-based development, algorithm validation, and performance analysis of the proposed technologies ● Technical guidance and collaboration with the industrial partner to support system integration and experimental evaluation
Expected R&R of Korean SME Partner	The Korean SME partner will be responsible for the development, integration, and validation of the software prototype and integration with hardware systems required for the proposed anomaly detection technology. The SME will play a key role in translating AI algorithms into a practical industrial setting. The specific roles and responsibilities of the Korean SME partner include: ● Design and development of data acquisition hardware, including sensor systems, data transmission, and data analysis platform ● System integration of analytical algorithms with hardware systems ● Execution of laboratory-scale and real-world experiments, including data collection, testing, and performance evaluation ● Contribution to technology refinement and commercialization planning, leveraging industrial expertise and manufacturing capabilities

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	The proposed joint R&D project is expected to deliver an end-to-end analytical anomaly detection prototype capable of quick and resource efficient detection in real-world industrial environments. The project will advance the state of the art in AI-based anomaly detection, adaptive sampling, and anomaly mitigation support systems, achieving measurable performance improvements. From an industrial perspective, this project will provide the Korean SME partner with core AI-enabled anomaly detection and monitoring resource management technologies in general industrial settings. The outcomes will support **technology commercialization and product development** to enable cost-effective anomaly detection solutions for applications like advanced manufacturing and Internet of Things. More broadly, the project is expected to strengthen the competitiveness of Korean SMEs and contribute to the growth of the AI-driven production industry through international collaboration.

6. References

(1) Xian, X., Wang, A., & Liu, K. (2018). A nonparametric adaptive sampling strategy for online monitoring of big data streams. Technometrics, 60(1), 14-25.

(2) Wang, A., Xian, X., Tsung, F., & Liu, K. (2018). A spatial-adaptive sampling procedure for online monitoring of big data streams. Journal of Quality Technology, 50(4), 329-343.

(3) Zan, X., Wang, D., & Xian, X. (2023). Spatial rank-based augmentation for nonparametric online monitoring and adaptive sampling of big data streams. Technometrics, 65(2), 243-256.

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

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